
CRZ42 G2 Trouble Shooting And FAQ List

Contents

CRZ42 G2 Trouble Shooting And FAQ List	1
PMU2 - Overtemperature-Level1	6
PMU10 - PMU loop fault	6
PMU11 - PMU minor fault	6
PMU12 - PMU major fault	6
PMU13 - PMU no available battery	9
PMU35 - Overtemperature-Level2	10
PMU36 - Undervoltage	10
PMU37 - Overvoltage	10
PMU38 - Power supply output failure	10
PMU41 - No battery pack available	10
PMU46 - Relay (MOS) error	10
PMU47 - Pre-charge error	10
PMU48 - Pre-charge hardware error	11
PMU49 - Negative MOSFET temperature sensor error	11
PMU52 - Current sensor error	11
PMU57 - KSI Pre-MOS error	11
PMU59 - KSI MOS error	11
PMU61 - Battery pack 1 does not match	11
PMU62 - Battery pack 2 does not match	11
PMU63 - Battery pack 3 does not match	11
PMU64 - Battery pack 4 does not match	12
PMU65 - Battery pack 5 does not match	12
PMU66 - Battery pack 6 does not match	12
PMU67 - Abnormal charging status	12
TR2/TL2 - Drive motor stalled	12
TR4/TL4 - Motor overspeed protection	14
TR5/TL5 - Motor encoder error	14
TR6/TL6 - Controller phase loss	14
TR7/TL7 - MOSFET error	15
TR8/TL8 - Controller undervoltage	15

TR9/TL9 - Controller overvoltage	15
TR12/TL12 - Motor overtemperature	15
TR13/TL13 - Controller overtemperature	16
TR14/TL14 - Software Overcurrent	16
TR15/TL15 - Hardware Overcurrent	16
TR16/TL16 - Potentiometer error	17
TR19/TL19 - Operating sequence error	17
TR22/TL22 - Electromagnetic valve error	17
TR23/TL23 - Battery compartment (PMU)communication error	19
TR24/TL24 - Battery compartment (PMU) shutdown	20
TR25/TL25 - CAN timeout error - left drive	22
TR26/TL26 - CAN timeout error - right drive	22
TR27/TL27 - CAN timeout error - left blade	23
TR29/TL29 - CAN timeout error - right blade	24
TR30/TL30 - Power off warning	25
TR31/TL31 - Seat switch verification error	25
TR32/TL32 - Software authentication error	26
MR1/ML1 - Motor positioning error	26
MR2/ML2 - Drive motor stalled	27
MR3/ML3 - Blade motor low speed	27
MR4/ML4 - Blade motor over speed	27
MR6/ML6 - Controller phase loss	28
MR7/ML7 - MOSFET error	28
MR8/ML8 - Controller undervoltage	28
MR9/ML9 - Controller overvoltage	28
MR13/ML13 - Controller overtemperature	29
MR14/ML14 - Software overcurrent	29
MR15/ML15 - Hardware overcurrent	29
MR19/ML19 - Operating sequence error	29
MR25/ML25 - CAN timeout error - left drive	30
MR26/ML26 - CAN timeout error - right drive	31

MR27/ML27 - CAN timeout error - left blade	32
MR29/ML29 - CAN timeout error - right blade	33
MR31/ML31 - Blade switch verification error	34
MR32/ML32 - Software authentication error	35
D1 - CAN timeout with PMU	36
D2 - CAN timeout with left traction controller	37
D3 - CAN timeout with right traction controller	38
D4 - CAN timeout with left blade controller	39
D6 - CAN timeout with right blade controller	40

Tools: Computer、CAN DEBUG TOOL (ERP: R0203845-00) 、16-Pin adapter (ERP: R0203075-00) 、Jack、Multimeter.



CAN DEBUG TOOL (ERP: R0203845-00) | 16-Pin adapter (ERP: R0203075-00)



Jack



Multimeter

PMU2 - Overtemperature-Level1

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU10 - PMU loop fault

At least one channel in the battery compartment is unable to discharge, but the performance will not decrease, vehicle still can be worked.

If you feel a decrease in mowing performance with this fault code, the battery compartment should be replaced.

PMU11 - PMU minor fault

There have occurred a over-current(Level 1) or over-temperature(Level 1) fault. The vehicle's performance will be limited with this code, but this fault can be restored, no need to pay attention.

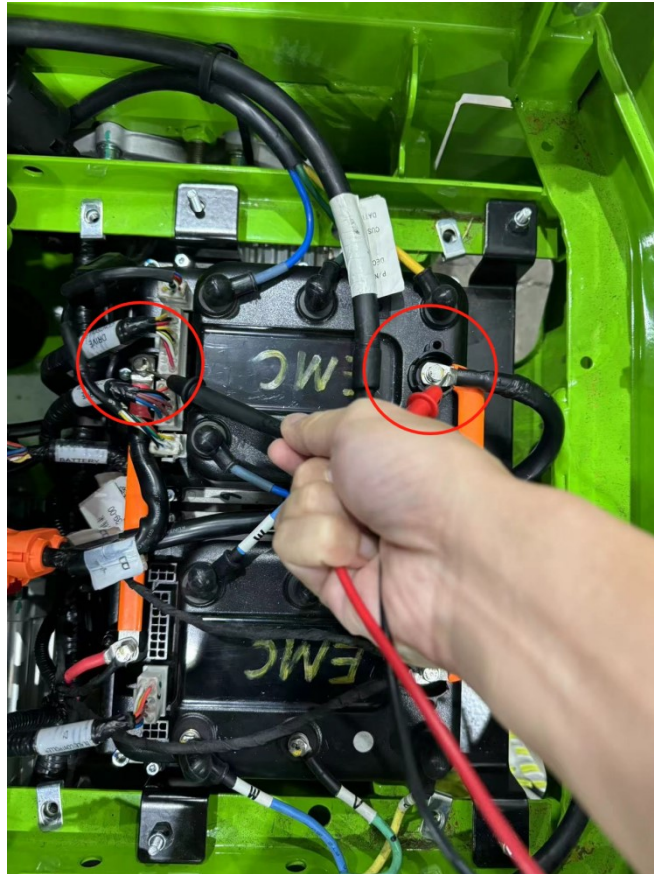
PMU12 - PMU major fault

Check: Use "ToolsForCAN" to identify the detail fault -

1. "PreCharge Function Error" of PMU12

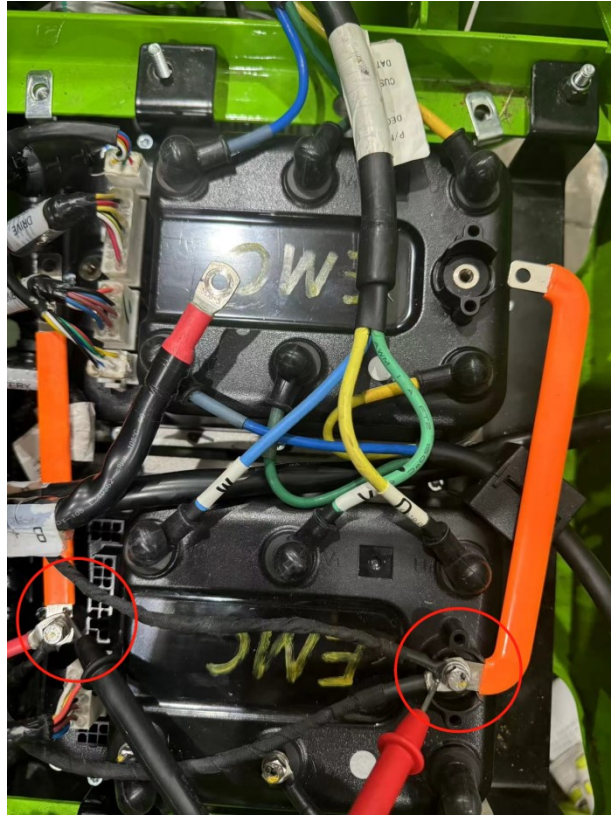
PMU FAULT LIST PART1			
Function	Status		
Avg Curr Overcurr	☒ 0: Normal	☐ 1: Avg Curr L1	
	☐ 2: Avg Curr L2	☐ 3: Spare	
PMUNeg. MOSOvertemp	☒ 0: Normal	☐ 1: OvertempL1	
	☐ 2: OvertempL2	☐ 3: NTCErr	
Peak Curr Overcurr	☒ 0: Normal	☐ 1: Error	
Feedback Curr Overcurr	☒ 0: Normal	☐ 1: Error	
PreCharge Function	☐ 0: Normal	☒ 1: Error	
PMU Software Auth	☒ 0: Normal	☐ 1: Error	
PMU Undervoltage Error	☒ 0: Normal	☐ 1: Error	
PMU Overvoltage Error	☒ 0: Normal	☐ 1: Error	

- ① Adjust the dial position of a multimeter to the diode position to measure the diode value of the traction controller's B+ and B- as shown in the picture below(Red lead to black wire, black lead to red wire).
 - a. If the value is approximately equal to $0.7 \pm 0.2V$, you should replace a new battery box.
 - b. Otherwise follow the steps.



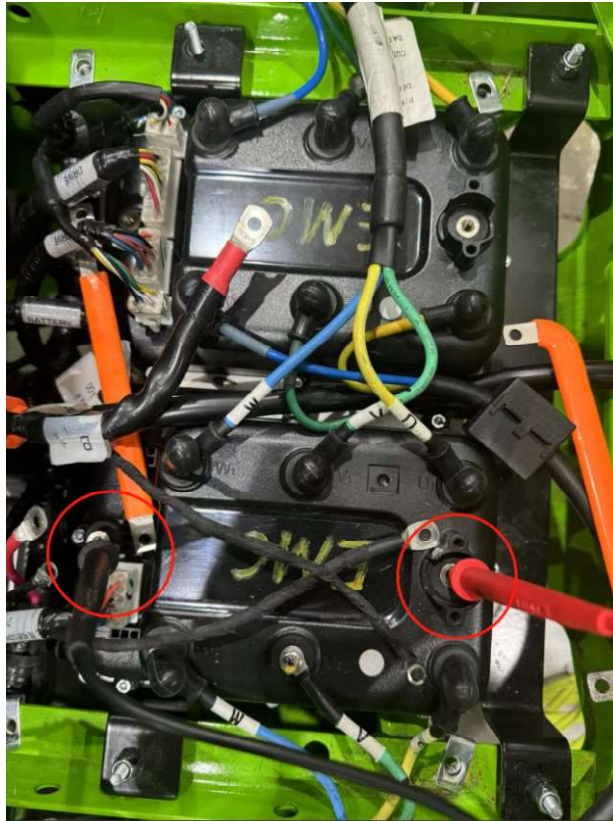
Correct value should be approximately equal to $0.7 \pm 0.2V$

- ② Disconnect the traction controller's B+ and B- wires and use a multimeter to measure the diode value as shown in the picture below (Red lead to negative electrode, black lead to positive electrode).
- If the value is approximately equal to $0.7 \pm 0.2V$, you should replace traction controller.
 - Otherwise follow the steps.



Correct value should be approximately equal to $0.7 \pm 0.2V$

- ③ Disconnect the blade controller's B+ and B- wires and use a multimeter to measure the diode value of the blade controller as the picture showed below (Red lead to negative electrode, black lead to positive electrode).
- If the value is approximately equal to $0.7 \pm 0.2V$, you should replace the DC-DC.
 - Otherwise replace the blade controller.



Correct value should be approximately equal to $0.7 \pm 0.2V$

2. Other errors of PMU12

Replace the battery compartment.

PMU13 - PMU no available battery

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU35 - Overtemperature-Level2

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the battery compartment.

PMU36 - Undervoltage

1. Charge the batteries.
2. Make sure the battery compartment on the correct platform is used.
3. Replace the battery compartment.

PMU37 - Overvoltage

1. Make sure the battery compartment on the correct platform is used.
2. Replace the battery compartment.

PMU38 - Power supply output failure

Disconnect the power harness from battery compartment:

1. If the fault code disappeared, follow the "PMU12" trouble shooting guide to find the detail issue.
2. Otherwise, replace the battery compartment.

PMU41 - No battery pack available

1. Make sure that the inserted batteries are correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU46 - Relay (MOS) error

Replace the battery compartment.

PMU47 - Pre-charge error

Follow the "PreCharge Function Error" of PMU12 trouble shooting guide to find the detail issue.

PMU48 – Pre-charge hardware error

Replace the battery compartment.

PMU49 – Negative MOSFET temperature sensor error

Replace the battery compartment.

PMU52 – Current sensor error

Replace the battery compartment.

PMU57 – KSI Pre-MOS error

Replace the battery compartment.

PMU59 – KSI MOS error

Replace the battery compartment.

PMU61 – Battery pack 1 does not match

1. Make sure that the inserted battery in cabin 1 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU62 – Battery pack 2 does not match

1. Make sure that the inserted battery in cabin 2 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU63 – Battery pack 3 does not match

1. Make sure that the inserted battery in cabin 3 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU64 - Battery pack 4 does not match

1. Make sure that the inserted battery in cabin 4 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU65 - Battery pack 5 does not match

1. Make sure that the inserted battery in cabin 5 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU66 - Battery pack 6 does not match

1. Make sure that the inserted battery in cabin 6 is correct, not mixed with different voltage platform.
2. Replace the battery compartment.

PMU67 - Abnormal charging status

1. Check if the charger's input plug is properly plugged in.
2. Check if the vehicle can be powered on and worked properly:
 - ① Can't be powered on and worked, then replace the battery compartment.
 - ② Otherwise, replace the charge socket.

TR2/TL2 - Drive motor stalled

1. TR2
 - ① Turn off the power, use a jack to lift the vehicle's rear wheels, disengage the manual parking brake, try to rotate the rear wheel of right side.
 - a. If the wheel can't be rotated, replace the right traction motor.
 - b. Otherwise follow the steps below.



② Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR2 fault code, replace the traction controller.
- b. If fault code change to TL2, you should replace the right traction motor.

2. TL2

① Turn off the power, use a jack to lift the vehicle's rear wheels, disengage the manual parking brake, try to rotate the rear wheel of left side.

- a. If the wheel can't be rotated, replace the left traction motor.
- b. Otherwise follow the steps below.



② Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TL2 fault code, replace the traction controller.
- b. If fault code change to TR2, you should replace the left traction motor.

TR4/TL4 - Motor overspeed protection

1. TR4

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR4 fault code, replace the traction controller.
- b. If fault code change to TL4, you should replace the right traction motor.

2. TL4

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TL4 fault code, replace the traction controller.
- b. If fault code change to TR4, you should replace the left traction motor.

TR5/TL5 - Motor encoder error

1. TR5

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR6 fault code, replace the traction controller.
- b. If fault code change to TL6, you should replace the right traction motor.

2. TL5

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TL6 fault code, replace the traction controller.
- b. If fault code change to TR6, you should replace the left traction motor.

TR6/TL6 - Controller phase loss

1. TR6

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR6 fault code, replace the traction controller.

b. If fault code change to TL6, you should replace the right traction motor.

2. TL6

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

a. If still have TL6 fault code, replace the traction controller.

b. If fault code change to TR6, you should replace the left traction motor.

TR7/TL7 - MOSFET error

Replace the traction controller.

TR8/TL8 - Controller undervoltage

- ① Charge the batteries.
- ② Make sure the controller on the correct platform is used.
- ③ Flash the right firmware to the traction controller. 60V and 80V platform have different firmware.
- ④ Replace the traction controller.

TR9/TL9 - Controller overvoltage

- ① Make sure the controller on the correct platform is used.
- ② Flash the right firmware to the traction controller. 60V and 80V platform have different firmware.
- ③ Replace the traction controller.

TR12/TL12 - Motor overtemperature

1. Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, follow the steps below.

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

① TR12

a. If still have TR12 fault code, replace the traction controller.

b. If fault code change to TL12, you should replace the left traction motor.

② TL12

- a. If still have TL12 fault code, replace the traction controller.
- b. If fault code change to TR12, you should replace the right traction motor.

TR13/TL13 - Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the traction controller.

TR14/TL14 - Software Overcurrent

1. TR14

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR14 fault code, replace the traction controller.
- b. If fault code change to TL14, you should replace the right traction motor.

2. TL14

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TL14 fault code, replace the traction controller.
- b. If fault code change to TR14, you should replace the left traction motor.

TR15/TL15 - Hardware Overcurrent

1. TR15

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TR15 fault code, replace the traction controller.
- b. If fault code change to TL15, you should replace the right traction motor.

2. TL15

Swap the M5 terminals of left and right traction motors' phase harness(Totally six screws) and swap the 6 pins encoder sensor connectors.

- a. If still have TL15 fault code, replace the traction controller.

b. If fault code change to TR15, you should replace the left traction motor.

TR16/TL16 – Potentiometer error

Check: Use “ToolsForCAN” to read the value of right/left potentiometer -

- ① If the value is larger than 4.6 or smaller than 0.2, replace the right/left potentiometer.
- ② Replace the traction controller.
- ③ Replace the control harness.

TR19/TL19 – Operating sequence error

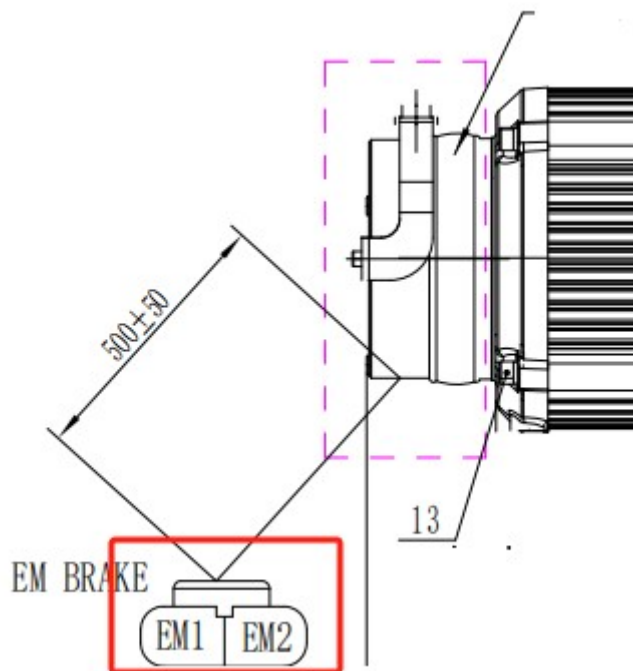
Check: Use “ToolsForCAN” to identify the status of switches and potentiometer -

Make sure there have no operations before sitting on the seat. The status should meet the table below before operating.

Switch & Controllers		State
Seat Switch		ON
Left Parking Switch		ON
Right Parking Switch		ON
PTO Switch		OFF
Left Potentionmeter	2.40	2.3~2.5V
Right Potentionmeter	2.40	2.3~2.5V
Left Traction Controller		No Fault
Right Traction Controller		No Fault
PMU		No Fault

TR22/TL22 – Electromagnetic valve error

Measure the resistance of the right/left electromagnetic brake.



1. The value is within the range(40~200Ω).

Swap the left and right traction motors' EM brake connectors.

① If still have TR22/TL22 fault code

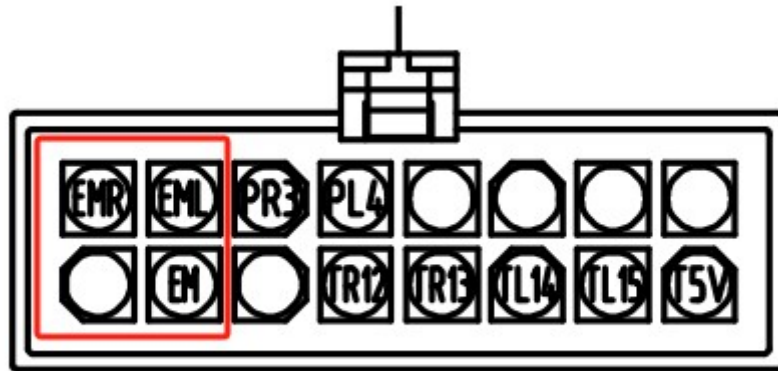
a. Check if the pins of the traction controller are bent or retracted, if can be fixed, try to fix them.

b. Check if the pins of the connector of the harness side are bent or retracted, if can be fixed, try to fix them, otherwise, replace the control harness.

供应商	名称	型号
TE	外壳	282104-1
	端子	282404-1
	单线密封	281934-4

供应商	名称	型号
TE	外壳	282104-1
	端子	282404-1
	单线密封	281934-4





DRIVE CONTROLLER		
供应商	名称	型号
TE	外壳	794824-1
	端子	770904-1
	接口密封	1-1586362-6

打胶

c. Replace the traction controller.

② If TR22/TL22 disappeared and get a TL22/TR22 fault code, replace the right/left traction motor.

2. The value is not within the range(40~200Ω).

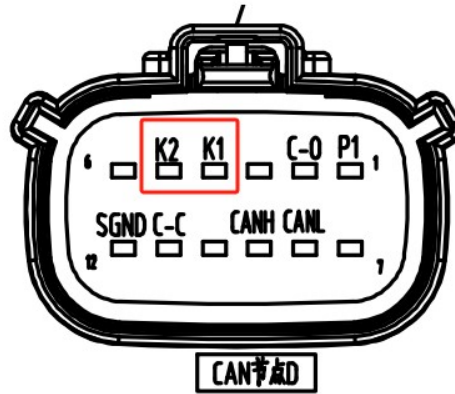
Replace the right/left traction motor.

TR23/TL23 - Battery compartment (PMU)communication error

1. Check the pins of connector of the battery compartment side:

a. If the pins are bent or retracted, if can be fixed, try to fix them.

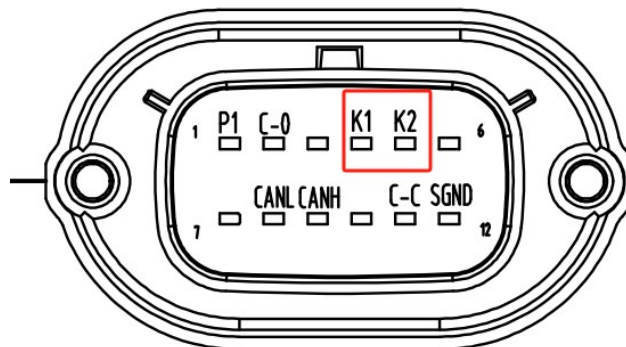
b. Replace the battery compartment.



BATTERY 12 LOGIC		
供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

2. Check the pins of the connector of the harness side:
 - a. If the pins bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

供应商	名称	型号
MOLEX	外壳	477259010
	端子	330000002

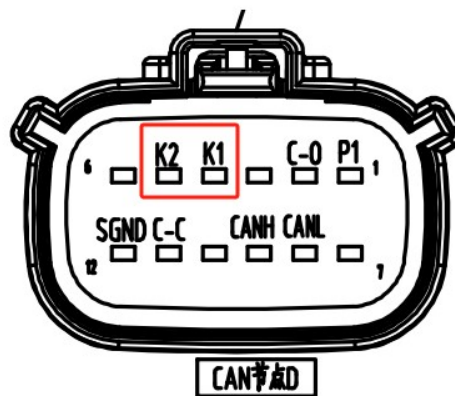


TR24/TL24 - Battery compartment (PMU) shutdown

1. If you turn off the vehicle, you will get this fault code, it's normal.
2. Otherwise:
 - ① Check if the connector of the key switch is connected well.

② Check the pins of connector of the battery compartment side:

- a. If the pins are bent or retracted, if can be fixed, try to fix them.
- b. Replace the battery compartment.

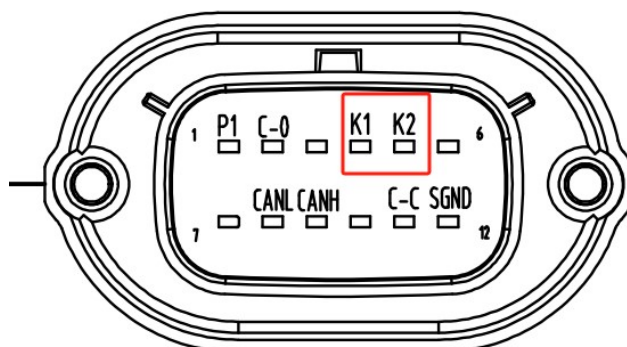


BATTERY 12 LOGIC		
供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

③ Check the pins of the connector of the harness side:

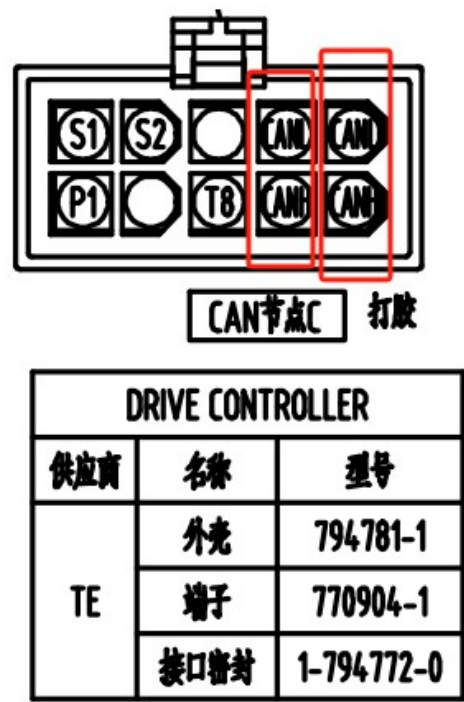
- a. If the pins are bent or retracted, if can be fixed, try to fix them.
- b. Replace the control harness.

供应商	名称	型号
MOLEX	外壳	477259010
	端子	330000002



TR25/TL25 - CAN timeout error - left drive

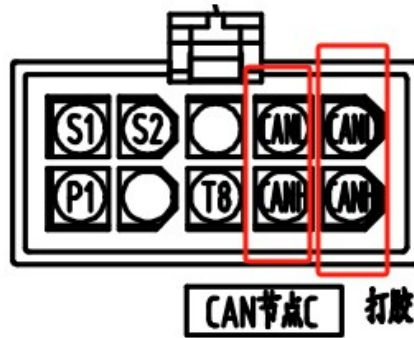
Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.



- ① If the resistance is around 60 Ohms(CANH to CANL):
 - a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the traction controller.
- ② Otherwise:
 - a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

TR26/TL26 - CAN timeout error - right drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

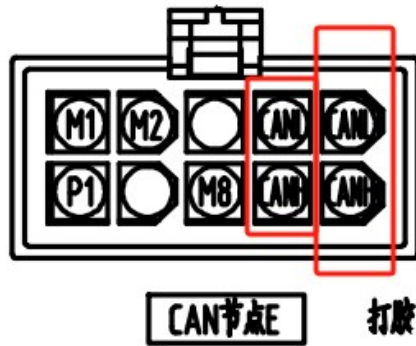


DRIVE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

- ① If the resistance is around 60 Ohms(CANH to CANL):
 - a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the traction controller.
- ② Otherwise:
 - a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

TR27/TL27 - CAN timeout error - left blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

① If the resistance is around 60 Ohms(CANH to CANL):

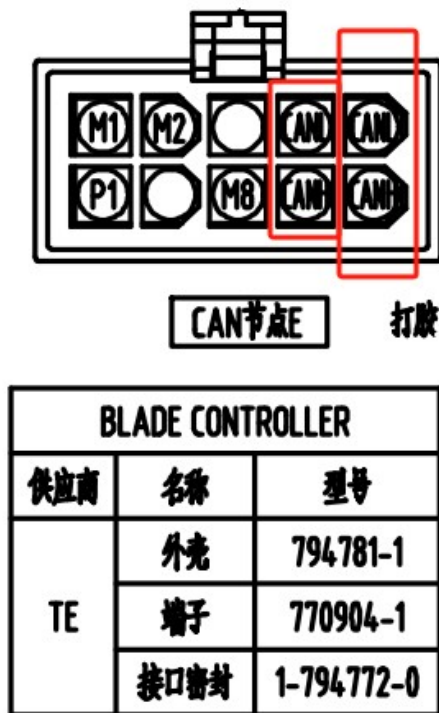
- a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
- b. Replace the blade controller.

② Otherwise:

- a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
- b. Replace the control harness.

TR29/TL29 - CAN timeout error - right blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



① If the resistance is around 60 Ohms(CANH to CANL):

- a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
- b. Replace the blade controller.

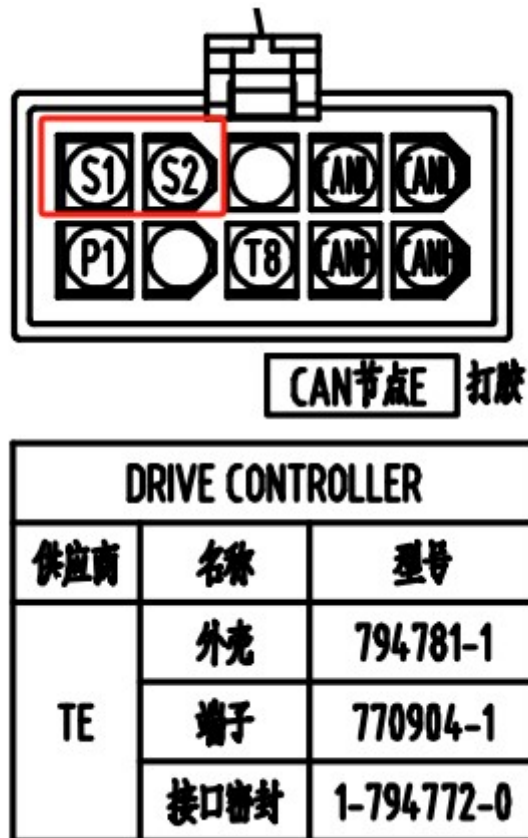
② Otherwise:

- a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
- b. Replace the control harness.

TR30/TL30 - Power off warning

TR31/TL31 - Seat switch verification error

1. Check if the connector of the seat switch is connected well.
2. Check the pins of connector of the traction controller's side:
 - a. If the pins of the traction controller are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the traction controller.
3. Check the pins of the connector of the harness side:
 - a. If the pins of are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.



4. Replace the seat switch.

TR32/TL32 - Software authentication error

Replace the traction controller.

MR1/ML1 - Motor positioning error

1. MR1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have MR1 fault code, replace the blade controller.
- b. If fault code change to ML1, you should replace the right blade motor.

2. ML1

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have ML1 fault code, replace the blade controller.
- b. If fault code change to MR1, you should replace the left blade motor.

MR2/ML2 – Drive motor stalled

1. MR2

- ① Turn off the power, try to rotate the right side blade.
 - a. If the blade can't be rotated or very tight, replace the right blade motor.
 - b. Otherwise follow the steps below.
- ② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).
 - c. If still have MR2 fault code, replace the blade controller.
 - d. If fault code change to ML2, you should replace the right blade motor.

2. ML2

- ① Turn off the power, try to rotate the left side blade.
 - a. If the blade can't be rotated or very tight, replace the left blade motor.
 - b. Otherwise follow the steps below.
- ② Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).
 - c. If still have ML2 fault code, replace the blade controller.
 - d. If fault code change to MR2, you should replace the left blade motor.

MR3/ML3 – Blade motor low speed

1. MR3

- Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).
- a. If still have MR3 fault code, replace the blade controller.
 - b. If fault code change to ML3, you should replace the right blade motor.

2. ML3

- Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).
- a. If still have ML3 fault code, replace the blade controller.
 - b. If fault code change to MR3, you should replace the left blade motor.

MR4/ML4 – Blade motor over speed

1. MR4

- Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have MR4 fault code, replace the blade controller.
- b. If fault code change to ML4, you should replace the right blade motor.

2. ML4

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have ML4 fault code, replace the blade controller.
- b. If fault code change to MR4, you should replace the left blade motor.

MR6/ML6 - Controller phase loss

1. MR6

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have MR6 fault code, replace the blade controller.
- b. If fault code change to ML6, you should replace the right blade motor.

2. ML6

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have ML6 fault code, replace the blade controller.
- b. If fault code change to MR6, you should replace the left blade motor.

MR7/ML7 - MOSFET error

Replace the blade controller.

MR8/ML8 - Controller undervoltage

- ① Charge the batteries.
- ② Make sure the controller on the correct platform is used.
- ③ Flash the right firmware to the blade controller. 60V and 80V platform have different firmware.
- ④ Replace the blade controller.

MR9/ML9 - Controller overvoltage

- ① Make sure the controller on the correct platform is used.
- ② Flash the right firmware to the blade controller. 60V and 80V platform have different firmware.

- ③ Replace the blade controller.

MR13/ML13 – Controller overtemperature

Stop the vehicle for a period of time to allow it to cool down, if the fault still have after restarting, replace the blade controller.

MR14/ML14 – Software overcurrent

1. MR14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have MR14 fault code, replace the blade controller.
- b. If fault code change to ML14, you should replace the right blade motor.

2. ML14

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have ML14 fault code, replace the blade controller.
- b. If fault code change to MR14, you should replace the left blade motor.

MR15/ML15 – Hardware overcurrent

1. MR15

Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have MR15 fault code, replace the blade controller.
- b. If fault code change to ML15, you should replace the right blade motor.

2. ML15

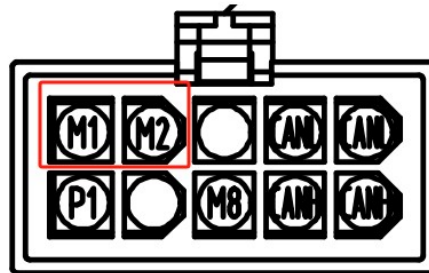
Swap the M5 terminals of left and right blade motors' phase harness(Totally six screws).

- a. If still have ML15 fault code, replace the blade controller.
- b. If fault code change to MR15, you should replace the left blade motor.

MR19/ML19 – Operating sequence error

1. Make sure the blade switch is on the "OFF" position, then try to turn on the blade again.
2. Check if the connector of the blade switch is connected well.

3. Check the pins of connector of the blade controller's side:
 - a. If the pins of the blade controller are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the blade controller.
4. Check the pins of the connector of the harness side:
 - a. If the pins of are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.



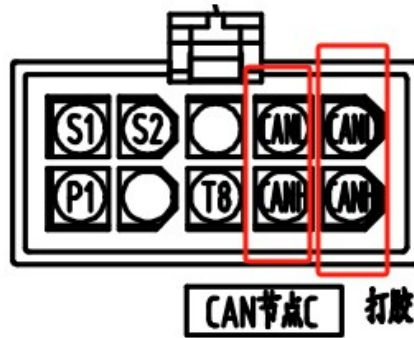
CAN节点 打胶

BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

5. Replace the blade switch,

MR25/ML25 - CAN timeout error - left drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

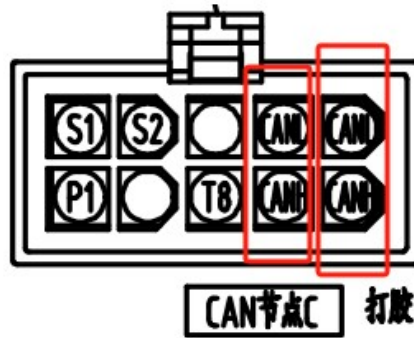


DRIVE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

- ① If the resistance is around 60 Ohms(CANH to CANL):
 - a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the traction controller.
- ② Otherwise:
 - a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

MR26/ML26 - CAN timeout error - right drive

Use multimeter to measure the resistance from the traction controller's 10 pins connector as the picture below.

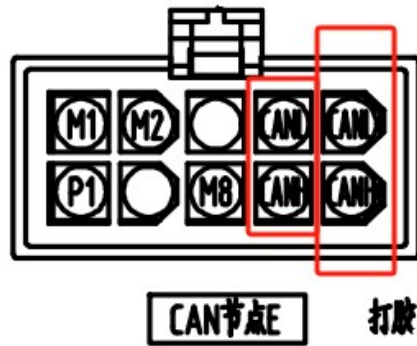


DRIVE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

- ① If the resistance is around 60 Ohms(CANH to CANL):
 - a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the traction controller.
- ② Otherwise:
 - a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

MR27/ML27 - CAN timeout error - left blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.

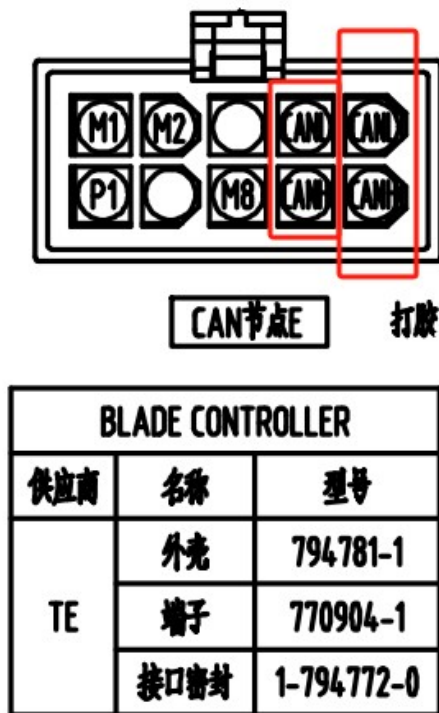


BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

- ① If the resistance is around 60 Ohms(CANH to CANL):
 - a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the blade controller.
- ② Otherwise:
 - a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.

MR29/ML29 - CAN timeout error - right blade

Use multimeter to measure the resistance from the blade controller's 10 pins connector as the picture below.



① If the resistance is around 60 Ohms(CANH to CANL):

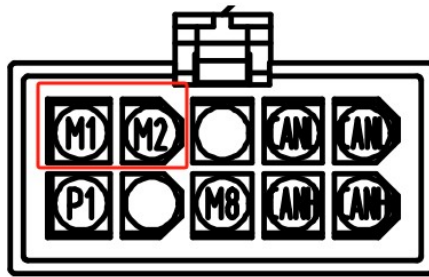
- a. Check if the pins of connector of controller side are bent or retracted, if can be fixed, try to fix them.
- b. Replace the blade controller.

② Otherwise:

- a. Check if the pins of the connector are bent or retracted, if can be fixed, try to fix them.
- b. Replace the control harness.

MR31/ML31 - Blade switch verification error

1. Check if the connector of the blade switch is connected well.
2. Check the pins of connector of the blade controller's side:
 - a. If the pins of the blade controller are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the blade controller.
3. Check the pins of the connector of the harness side:
 - a. If the pins of are bent or retracted, if can be fixed, try to fix them.
 - b. Replace the control harness.



CAN节点C 打胶

BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

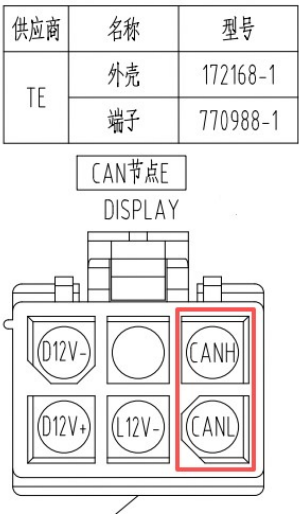
4. Replace the blade switch,

MR32/ML32 - Software authentication error

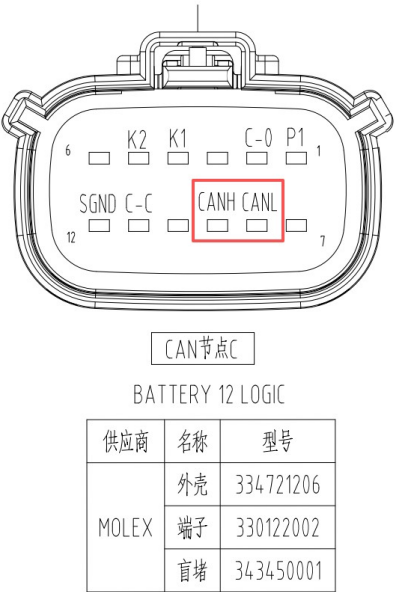
Replace the blade controller.

D1 - CAN timeout with PMU

- 1. If this fault exists simultaneously with any of D2/D3/D4/D5/D6, follow the steps below.
 - a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



- 2. If this fault exists alone without D2/D3/D4/D5/D6, follow the steps below.
 - a. Disconnect the connector of battery component side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the battery component.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

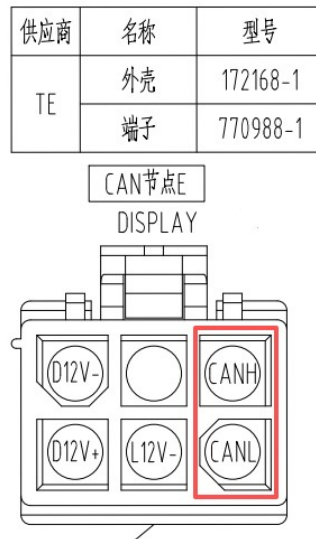


CAN节点C
BATTERY 12 LOGIC

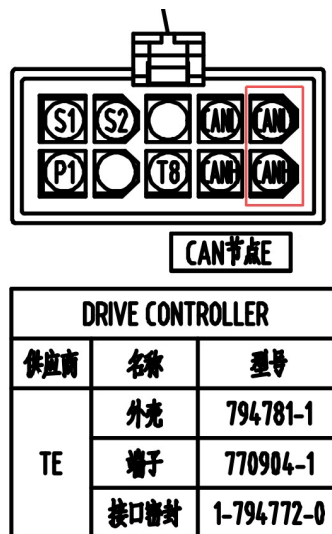
供应商	名称	型号
MOLEX	外壳	334721206
	端子	330122002
	盲堵	343450001

D2 - CAN timeout with left traction controller

1. If this fault exists simultaneously with D1, follow the steps below.
 - a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



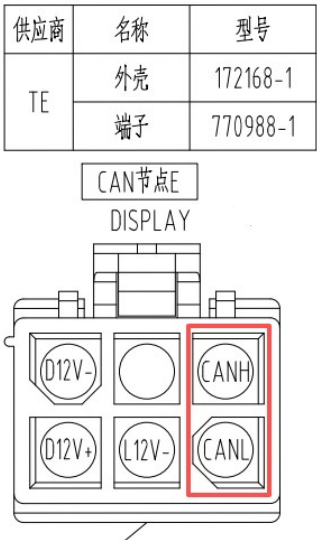
2. If this fault exists without D1, follow the steps below.
 - a. Disconnect the connector of traction controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



D3 – CAN timeout with right traction controller

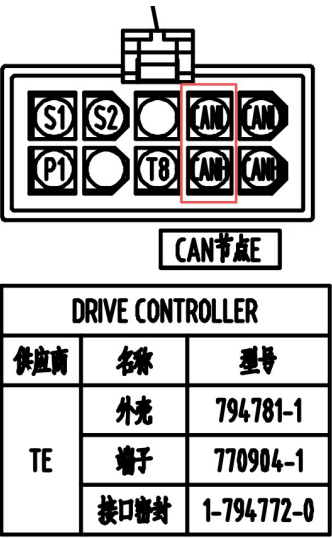
1. If this fault exists simultaneously with D1, follow the steps below.

- a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.
- b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



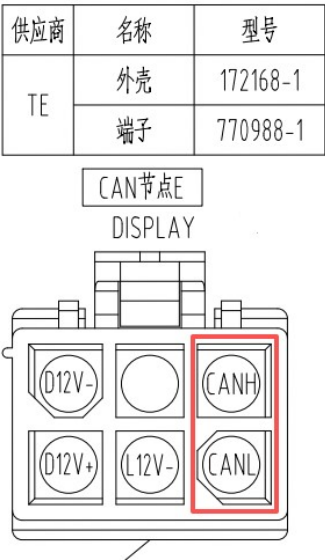
2. If this fault exists without D1, follow the steps below.

- a. Disconnect the connector of traction controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.
- b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

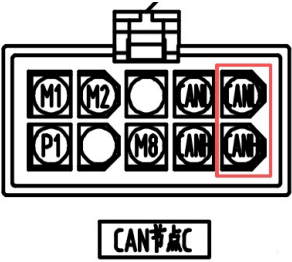


D4 - CAN timeout with left blade controller

- 1. If this fault exists simultaneously with D1, follow the steps below.
 - a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



- 1. If this fault exists without D1, follow the steps below.
 - a. Disconnect the connector of blade controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.
 - b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



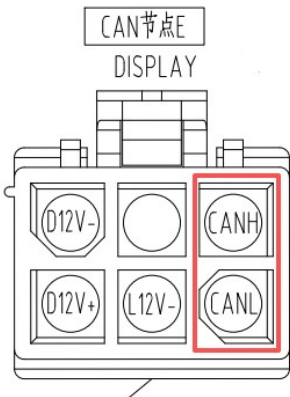
BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0

D6 – CAN timeout with right blade controller

1. If this fault exists simultaneously with D1, follow the steps below.

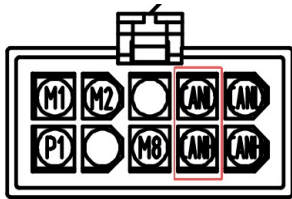
- a. Disconnect the connector of display, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the display.
- b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.

供应商	名称	型号
TE	外壳	172168-1
	端子	770988-1



2. If this fault exists without D1, follow the steps below.

- a. Disconnect the connector of blade controller side, use a multimeter to measure the resistance from “CANH” and “CANL”, if the value is approximately equal to 60Ω, replace the controller.
- b. Otherwise, the “CANH” and “CANL” of the control harness should be damaged, please replace a new control harness.



CAN节点C

BLADE CONTROLLER		
供应商	名称	型号
TE	外壳	794781-1
	端子	770904-1
	接口密封	1-794772-0